

PDA PRACTICE SHEET

AUTOMATA & COMPUTABILITY (CSE331)

1. $\{w \in \{0,1\}^*: 0^n 1^n \text{ where } n \geq 0\}$ [Spring24:B]
2. $\{w \in \{0,1\}^*: 0^n 1^m \text{ where } n \neq m\}$ [Spring24:B]
3. $\{w \in \{0,1\}^*: 0^n 1^{2n} \text{ where } n \geq 0\}$
4. $\{w \in \{0,1\}^*: 0^{3n} 1^{2n} \text{ where } n \geq 0\}$
5. $\{w \in \{0,1,2\}^*: 0^i 1^j 2^k \text{ where } i + j = k\}$
6. $\{w \in \{0,1,2\}^*: 0^i 1^j 2^k \text{ where } j = k + i\}$
7. $\{w \in \{0,1,2,3\}^*: 0^i 1^j 2^k 3^l\}$
8. $\{w \text{ is a palindrome}\}$ [Summer24:A|Odd Length][Summer24:B|Even Length]
9. $\{w^1 \# w^2 : \text{not of 0's in } w^1 = \text{not of 1's in } w^2\}$
10. $\{w \text{ starts and ends with same symbols}\}$ [Spring24:A] [Spring23:A: Eq 10=01|A: Eq 01=10]
11. $\{w \text{ odd length and in middle 0}\}$
12. $\{w \in \{0,1,2\}^*: 0^i 1^j 2^k \text{ where } i=j \text{ or } j=k\}$
13. $\{w \in \{0,1,2\}^*: 0^i 1^j 2^k \text{ where } i+j \geq k\}$
14. $L^1 L^2, L1 \rightarrow \text{odd 0's } L2 \rightarrow \text{Even 1's } |L^1| = |L^2|$
15. $\{a^n b^m c^k d^l e^k f^n, n \geq 0, m \geq 1 \text{ and } k \geq 2\}$
16. $\{w \in \{a,b,c\}^*: a^m b^n c^p \text{ where } n > m+p\}$
17. $\{w \in \{a,b,c\}^*: a^m b^n c^p \text{ where } n > m+p, m,n,p \geq 1\}$
18. $\{w \in \{a,b\}^*: a^m b^n \text{ where } m > 2n \text{ and } m,n \geq 1\}$
19. $\{w \in \{a,b\}^*: a^n b^m a^m b^m a^n, m,n \geq 1\}$
20. $\{w \in \{0,1\}^*: 0^n 1^n \text{ where } n \geq 2\}$
21. $\{w \in \{0,1\}^*: 0^m 1^n \text{ where } m \geq n\}$
22. $\{w \in \{0,1\}^*: 0^m 1^n \text{ where } m < n\}$
23. $\{w \in \{0,1\}^*: 0^n 0^{2n} 1^n\}$ [Fall22]
24. $\{w \in \{x,y\}^*: x^{2n} y^n, n \geq 1\}$
25. $\{w \in \{0,1\}^*: x^n x^m y^m z^n, n \geq 0\}$
26. $\{w^1 \# w^2 : \text{no of 00's in } w^1 = \text{no of 11's in } w^2\}$
27. $\{w \text{ length at least 2}\}$
28. $\{w \in \{(,)\}^*: w \text{ a valid parenthesis}\}$
29. $\{w \in \{0,1\}^*: 0^{n+2} 1^{3n}, n \geq 0\}$
30. $\{w \in \{x,y\}^*: x^n y^{n+1}, n \geq 1\}$
31. $\{w \in \{0,1\}^*: 0^n 1^n \text{ where } n \text{ is odd}\}$
32. $\{w \in \{0,1\}^*: 0^m 1^n \text{ where } m \geq n \text{ and } m, n \geq 1\}$
33. $\{w \in \{0,1\}^*: \text{the length of } w \text{ is divisible by 4}\}$ [Fall22]
34. $\{w \in \{0,1\}^*: \text{the number of 1's in } w \text{ is multiple of 3}\}$
35. $\{w \in \{0,1\}^*: \text{contains even no of 0's}\}$
36. $\{w \in \{0,1\}^*: \text{the parity of 0s and 1s are different in } w\}$
37. $\{w \in \{0,1\}^*: \text{the number of 0s and 1s are different in } w\}$ [Spring24:A]
38. $\{w \in \{0,1\}^*: 0^m 1^n \text{ where } m,n \geq 1 \text{ and } m > n+2\}$
39. $\{w \in \{0,1\}^*: ww^R : w^R \text{ is a reverse complement of } w\}$ [w=0100, w^R=1011] [Spring 22]
40. $\{w \in \{0,1\}^*: w_1 \# w_2 \# w_3 : \text{count function: } C(w_1,0)=C(w_2,0) \text{ or } C(w_2,1)=C(w_3,1)\}$
[Summer 24:A|B]
41. $\{w \text{ starts and ends with different symbols}\}$ [Spring24:B]
42. $\{w \in \{0,1\}^*: \text{contains at least two 1's}\}$ [Fall23]
43. $\{w \in \{0,1\}^*: 1^n 0^n 1^{2n} \text{ where } n \geq 0\}$ [Fall23]